Name-Kashish Chelwani

PRN-24070126082

AIML-B1

Theory of SVD(Singular Value Decomposition)

Singular Value Decomposition (SVD)

Singular Value Decomposition (SVD) is a **matrix factorization technique** used for dimensionality reduction, data compression, and feature extraction. It decomposes a high-dimensional dataset into smaller matrices that capture the most important patterns in the data.

SVD is widely used in **machine learning, image processing, recommender systems, natural language processing, and signal processing**.

The main objective of SVD is to **reduce the dimensionality of data while preserving the most important information**.

In high-dimensional datasets such as image datasets, many features may be redundant or correlated. SVD helps extract the most significant components of the data and represent them in a lower-dimensional space.

Mathematical Representation of SVD

For any matrix **A**, Singular Value Decomposition can be written as:

$$A = U\Sigma V^T$$

Where:

- **U** = Matrix containing left singular vectors
- **Σ (Sigma)** = Diagonal matrix containing singular values
- **V^T** = Transpose of matrix containing right singular vectors

Interpretation

- Singular values represent the **importance of each component**.
- Larger singular values indicate components that capture more information from the data.
- Smaller singular values correspond to less important components and can be removed during dimensionality reduction.

By selecting only the top **k singular values**, the dataset can be reduced from **high-dimensional space to k dimensions** while preserving most of the variance.

Key Concepts in SVD

1. Singular Values

Singular values represent the amount of **variance or information captured by each component**. Components with larger singular values contribute more to the data structure.

2. Dimensionality Reduction

In high-dimensional datasets, many features may be redundant or correlated. SVD reduces the number of dimensions by keeping only the most significant components.
For example:
Original dataset → 784 features
After SVD → 50 components
This significantly reduces computational complexity.

3. Variance Preservation

SVD preserves the most important information by retaining components that capture the highest variance in the dataset.
The cumulative explained variance helps determine the **optimal number of components (k)** required to retain most of the information.

4. Feature Transformation

SVD transforms the original features into a new set of orthogonal components that capture the underlying structure of the dataset.
These transformed features can then be used for **classification or visualization**.

Steps Involved in Applying SVD

1. Load the dataset and preprocess the data.
2. Standardize the features using feature scaling.
3. Apply Singular Value Decomposition to decompose the dataset into U, Σ, and V matrices.
4. Compute explained variance for each component.
5. Select top **k components** that capture most of the variance.
6. Transform the dataset into the reduced dimensional space.
7. Train a machine learning model using the reduced dataset.

Example of SVD

Consider an image dataset where each image is represented by **784 pixel values (28 × 28 image)**.
Each image is flattened into a vector of **784 features**, which forms a high-dimensional dataset.
Applying SVD:

Original feature space → 784 dimensions
After SVD → 50 components
Instead of using all 784 features, the model now uses only the **most informative components**.
This reduces computational complexity while maintaining important information from the original data.

Advantages of SVD

- Reduces dimensionality of datasets
- Removes redundant and correlated features
- Improves computational efficiency
- Helps mitigate the curse of dimensionality
- Preserves important information in the dataset
- Improves model training speed

Limitations of SVD

- Computationally expensive for extremely large datasets
- Choosing the optimal number of components can be challenging
- May remove useful information if too many components are discarded
- Does not directly consider class labels since it is an unsupervised technique

Result Analysis from Datasets

1. Digits Dataset

In the digits dataset, each image contains **64 pixel features** representing handwritten digits from 0 to 9.
Applying SVD reduced the dimensionality of the dataset by selecting the most important components while discarding less informative ones.

Classification Performance

Accuracy without SVD: 0.9629629629629629
Accuracy with SVD: 0.9388888888888889

Variance Analysis

The cumulative explained variance graph shows that a relatively small number of components capture most of the information in the dataset.
This means that many of the original features are redundant and can be removed without significantly affecting classification performance.

2. Fashion-MNIST Dataset

The Fashion-MNIST dataset contains **28 × 28 grayscale images**, resulting in **784 features per sample**.
This represents a much higher dimensional dataset compared to the digits dataset.

Applying SVD significantly reduces the dimensionality while preserving most of the variance in the data.

Classification Performance

Accuracy without SVD: : *0.8475238095238096*

Accuracy with SVD: *0.8213809523809524*

Dimensionality Reduction

Features before SVD: **784**

Features after SVD: **50 components**

Variance Analysis

The cumulative explained variance shows that most of the dataset's information is captured within the first few components.

This demonstrates that SVD effectively compresses high-dimensional image data.

Observations

From the experimental results, it can be observed that applying SVD helps reduce the dimensionality of image datasets while retaining most of the important information.

In the digits dataset, dimensionality reduction slightly improves computational efficiency while maintaining similar classification accuracy. In the Fashion-MNIST dataset, which has a much larger number of features, SVD significantly reduces the feature space and helps the model focus on the most important patterns in the data.

Overall, the experiment demonstrates that SVD is an effective dimensionality reduction technique for high-dimensional image datasets, helping mitigate the curse of dimensionality and improving the efficiency of machine learning models.